



**MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST
BENGAL (FORMERLY WBUT)**

School of Information Science and Technology

Department of Information Technology

END SEMESTER EXAMINATION

JAN - 2025

PAPER NAME: Advanced Computer Architecture

PROGRAM NAME: B.Tech IT

Time : 3 Hours

PAPER CODE: PEC-IT501B

SEMESTER: V

Full Marks : 70

7

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

GROUP - A

(Multiple Choice Type Questions)

Choose the correct alternatives of the following

10 x 1 = 10

MARKS CO PO BL

- i) Which of the following is NOT an architectural classification by Flynn's taxonomy?
 - a. SISD
 - b. SIMD
 - c. MISC
 - d. MISD
- ii) In which type of memory does data persist even when power is turned off?
 - a. Cache memory
 - b. Flash memory
 - c. Dynamic RAM
 - d. Registers
- iii) What is the primary purpose of a Translation Lookaside Buffer (TLB)?
 - a. To improve disk I/O speed
 - b. To map virtual addresses to physical addresses
 - c. To store cache coherence information
 - d. To reduce pipeline hazards
- iv) Which of the following is an advantage of RISC over CISC?
 - a. Higher instruction complexity
 - b. Simple hardware implementation
 - c. Smaller code size
 - d. Longer instruction execution time
- v) What is the term used for a processor's ability to execute more than one instruction per clock cycle?
 - a. Out-of-order execution
 - b. Superscalar processing
 - c. Hyper-threading
 - d. Multi-core processing
- vi) Which scheduling technique is most commonly used in dynamic instruction scheduling?
 - a. Static scheduling
 - b. Tomasulo's algorithm
 - c. Round-robin scheduling
 - d. Priority scheduling
- vii) What is the primary function of branch prediction in modern processors?
 - a. To execute parallel instructions
 - b. To handle interrupts efficiently

38

26

64

c. To reduce control hazards

d. To enhance memory access time

11. In vector processing, what is the term for the time it takes to start processing the first data element?

a. Latency

b. Throughput

c. Clock cycle

d. Vector stride

12. Which type of dependency occurs when one instruction depends on the result of a previous instruction?

a. Structural hazard

b. Data hazard

c. Control hazard

d. Memory hazard

13. What is the key feature of Harvard Architecture compared to Von Neumann Architecture?

a. Separate memory for instructions and data

b. Unified memory for instructions and data

c. Simpler control unit design

d. Higher latency for data transfer

GROUP - B

(Short Answer Type Questions)

Answer the following

2.a. Highlight the challenges associated with implementing parallel processing in modern architectures.

3 x 5 = 15			
MARKS	CO	PO	B
5	1	2	

OR

2.b. Describe the significance of Flynn's taxonomy in designing high-performance systems.

5	4	2
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3.a. Discuss the advantages of using a memory hierarchy in reducing the performance gap between CPU and memory.

5	5	3
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OR

3.b. Explain how scheduling enhances the efficiency of instruction execution in parallel processing systems.

5	6	6
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4.a. Identify the differences between instruction-level pipelining and arithmetic pipelining.

5	2	2
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OR

4.b. Why is cache coherence important in distributed memory architectures? Provide examples.

5	3	
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GROUP - C

(Long Answer Type Questions)

Answer the following

5.a.i. Define parallel processing and discuss its role in solving computationally intensive problems.

ii. Compare SIMD and MIMD architectures, focusing on their implementation challenges.

3 x 15 = 45		
MARKS	CO	
8	1	
7	1	

OR

5.b.i.	Explain the role of memory hierarchy in modern computer architectures with a detailed diagram.	8	5	5
ii.	How do static and dynamic scheduling techniques impact processor performance?	7	6	5
6.a.i.	Define pipelining and explain the stages of the classic RISC pipeline with a diagram.	8	2	5
ii.	Identify and discuss the types of pipeline hazards with solutions for overcoming them.	7	2	6
OR				
6.b.i.	Describe vector processing and its advantages in solving large-scale numerical problems.	8	1	4
ii.	What are the key features of vector processing that make it ideal for high-performance computing?	7	1	6
7.a.i.	Discuss the use of Petri nets in modeling computational tasks in parallel systems.	8	6	5
ii.	How does VLSI computation contribute to designing efficient parallel architectures?	7	1	6
OR				
7.b.i.	How does instruction pipelining contribute to increased throughput in processors?	8	2	6
ii.	Differentiate between vector processors and scalar processors with examples.	7	1	4